

A Uniform Subcase of ℓ_∞ -Stability for Quasi-ACK Structure

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Source question

The source is Geunsu Choi and Mingu Jung, *A generalized ACK structure and the denseness of norm attaining operators*, arXiv:2204.01991. In Remark 3.4, after proving that $c_0(Z)$ has quasi-ACK structure whenever Z has quasi-ACK structure, the paper states that it is unknown whether quasi-ACK structure is stable under ℓ_∞ -sum operations in general. The same difficulty is repeated after Theorem 4.6 for arbitrary families.

Corollary 3.3. *Let Z be a Banach space with quasi-ACK structure.*

- (a) *If K is a compact Hausdorff space, then $C(K, Z)$ has quasi-ACK structure.*
- (b) *If Ω is a completely regular Hausdorff space and if, in addition, Z is finite dimensional, then $C_b(\Omega, Z)$ has quasi-ACK structure.*
- (c) *$c_0(Z)$ has quasi-ACK structure.*

Proof. (a) is an obvious consequence of Theorem 3.1. For the item (b), note that $C_b(\Omega, Z)$ can be isometrically identified with $C(\beta\Omega, Z)$, where $\beta\Omega$ is the Stone-Ćech compactification of Ω . (c) follows from the observation that $c_0(Z)$ can be viewed as a closed subspace of $C(\beta\mathbb{N}, Z)$. $\square$

Remark 3.4. We do not know if the class of Banach spaces with quasi-ACK structure is stable under ℓ_∞ -sum operations in general. As far as we know, it is even unknown if property quasi- β is stable under ℓ_∞ -sums or not. Nevertheless, the assertion (b) of Corollary 3.3 yields that $\ell_\infty(Z)$ has quasi-ACK structure whenever Z is a finite dimensional Banach space with quasi-ACK structure.

In fact, Corollary 3.3 enables us to construct a non-trivial Banach space which is of quasi-ACK structure, but not ACK_p structure or property quasi- β . We first need the following technical lemma.

Lemma 3.5. *Let K be a compact Hausdorff space and Z be a Banach space. If $C(K, Z)$ has property B, then $C(K)$ has property B.*

Proof. Let X be a Banach space and $T \in \mathcal{L}(X, C(K))$ with $\|T\| = 1$. Fix $z_0 \in S_Z$ and define $\tilde{T} \in \mathcal{L}(X, C(K, Z))$ by

Definitions used in the packet

We use the terminology of the source. A quasi-ACK structure on a Banach space X consists of a 1-norming set $\Gamma \subseteq B_{X^*}$, a function $\rho : \Gamma \rightarrow [0, 1)$, and, for each $e^* \in \text{Ext}(B_{X^*})$, a subset $\Gamma_{e^*} \subseteq \Gamma$

such that

$$e^* \in \overline{\mathbb{T}\Gamma_{e^*}}^{w^*}, \quad \sup_{x^* \in \Gamma_{e^*}} \rho(x^*) < 1,$$

and the local ACK-type conditions (i)'–(vi)' of Definition 2.1 in the source hold on every nonempty relatively weak-star open subset of Γ_{e^*} .

For such fixed data, call

$$\Delta = \left(\bigcup_{e^* \in \text{Ext}(B_{X^*})} \Gamma_{e^*} \right) \setminus \{0\}$$

the active part of the quasi-ACK witness.

Result

Theorem 1. *Let $\{X_i\}_{i \in I}$ be a family of Banach spaces. Suppose that each X_i has quasi-ACK data $(\Gamma_i, \rho_i, \Gamma_{i,e^*})$, and put*

$$\Delta_i = \left(\bigcup_{e^* \in \text{Ext}(B_{X_i^*})} \Gamma_{i,e^*} \right) \setminus \{0\}.$$

Assume the active parts are uniformly bounded away from 1:

$$r := \sup_{i \in I} \sup_{x^* \in \Delta_i} \rho_i(x^*) < 1.$$

Then

$$Z = \left(\bigoplus_{i \in I} X_i \right)_{\ell_\infty}$$

has ACK_r structure. In particular, Z has quasi-ACK structure.

Idea of the proof

The source's obstruction for ℓ_∞ -sums is the lack of an explicit description of the extreme points of the dual ball. In the uniform subcase we do not need to describe them. Milman's converse to Krein-Milman shows that the union of all active quasi-ACK pieces is already norming. If the ρ -values on that union are uniformly below 1, the quasi-ACK local pieces can be used as ordinary ACK_r pieces.

Once each summand is upgraded this way, the ℓ_∞ -sum construction is coordinate-local: choose a point of the weak-star open set in one coordinate, apply the summand's ACK_r data there, and extend the resulting vector and operator by zero on all other coordinates.

Proof

Lemma 1. *Let X have quasi-ACK data $(\Gamma, \rho, \Gamma_{e^*})$, and let*

$$\Delta = \left(\bigcup_{e^* \in \text{Ext}(B_{X^*})} \Gamma_{e^*} \right) \setminus \{0\}.$$

If $r = \sup_{\Delta} \rho < 1$, then X has ACK_r structure with witnessing set Δ .

Proof. First Δ is 1-norming. Indeed, each extreme point $e^* \in \text{Ext}(B_{X^*})$ belongs to $\overline{\mathbb{T}\Gamma_{e^*}}^{w^*}$, hence to $\overline{\mathbb{T}\Delta}^{w^*}$. By Krein-Milman,

$$B_{X^*} = \overline{\text{co Ext}(B_{X^*})}^{w^*} \subseteq \overline{\text{aco}(\mathbb{T}\Delta)}^{w^*}.$$

The reverse inclusion is immediate from $\Delta \subseteq B_{X^*}$, so $B_{X^*} = \overline{\text{aco}(\mathbb{T}\Delta)}^{w^*}$, which is equivalent to Δ being 1-norming.

Let $\varepsilon > 0$ and let U be a nonempty relatively w^* -open subset of Δ . Pick $u^* \in U$. By definition of Δ , there is an extreme point $e_0^* \in \text{Ext}(B_{X^*})$ such that $u^* \in \Gamma_{e_0^*}$. Then

$$U_0 = U \cap \Gamma_{e_0^*}$$

is a nonempty relatively w^* -open subset of $\Gamma_{e_0^*}$. Apply the quasi-ACK condition to U_0 . It gives a nonempty $V \subseteq U_0$, a point $x_1^* \in V$, a vector $e \in S_X$, and an operator $F \in \mathcal{L}(X)$ satisfying the local conditions (i)'–(vi)' relative to the original set Γ . These same conditions hold a fortiori for the smaller norming set Δ . In condition (iv)' the bound is

$$|x^*(F(e))| \leq \rho(x_1^*) \leq r$$

for $x^* \in \Delta \setminus V_1$. Thus the ACK_r conditions hold for Δ . $\square$

We now prove Theorem 1. By Lemma 1, each X_i has ACK_r structure with the 1-norming set Δ_i . Let $P_i : Z \rightarrow X_i$ and $J_i : X_i \rightarrow Z$ denote the canonical projection and injection. Set

$$\Delta = \bigcup_{i \in I} P_i^*(\Delta_i) \subseteq B_{Z^*}.$$

This set is 1-norming for Z , because for $z = (z_i) \in Z$,

$$\sup_{\phi \in \Delta} |\phi(z)| = \sup_{i \in I} \sup_{x^* \in \Delta_i} |x^*(z_i)| = \sup_{i \in I} \|z_i\| = \|z\|.$$

Let U be a nonempty relatively w^* -open subset of Δ and fix $P_{i_0}^* x_0^* \in U$, with $x_0^* \in \Delta_{i_0}$. Then

$$U_{i_0} = \{x^* \in \Delta_{i_0} : P_{i_0}^* x^* \in U\}$$

is a nonempty relatively w^* -open subset of Δ_{i_0} . Apply ACK_r structure of X_{i_0} to U_{i_0} and ε , obtaining

$$V_{i_0} \subseteq U_{i_0}, \quad x_1^* \in V_{i_0}, \quad e_{i_0} \in S_{X_{i_0}}, \quad F_{i_0} \in \mathcal{L}(X_{i_0}).$$

Define

$$V = P_{i_0}^*(V_{i_0}), \quad z_1^* = P_{i_0}^* x_1^*, \quad e = J_{i_0} e_{i_0}, \quad F = J_{i_0} F_{i_0} P_{i_0}.$$

Then $\|F\| = \|F(e)\| = 1$, $z_1^*(F(e)) = 1$, and $F^*(z_1^*) = z_1^*$.

It remains only to check the three conditions involving all functionals in Δ . Let $z^* = P_j^* y^* \in \Delta$. If $j = i_0$, the assertions are exactly the ACK_r assertions for X_{i_0} . If $j \neq i_0$, then

$$z^*(F(e)) = 0, \quad F^*(z^*) = 0.$$

Hence condition (iv)' is automatic, and $\text{dist}(F^*(z^*), \text{aco}\{0, V\}) = 0$, which gives condition (v)'. Condition (vi)' concerns only the elements of $V = P_{i_0}^*(V_{i_0})$, so it also follows from the summand X_{i_0} . Therefore Z has ACK_r structure.

Limitations

This is a partial result, not a solution of the full source problem. The general case remains open when the available quasi-ACK witnesses are genuinely non-uniform, i.e. when the active ρ -values cannot be bounded away from 1. The result also does not settle the source's separate question about stability of property quasi- β under ℓ_∞ sums.

Novelty and literature check

The cheap run indexes were searched for arXiv:2204.01991, generalized ACK, quasi-ACK, property quasi- β , norm attaining operators, and ℓ_∞ -sum stability. No duplicate packet or attempt for this exact target was found. Web searches for the paper title, quasi-ACK structure, ℓ_∞ -sums, and property quasi- β found the source paper but no later resolution of the ℓ_∞ -stability question in this bounded pass.

Human-review recommendation

This packet is a likely valid partial result. Review should focus on Lemma 1, especially the use of Krein-Milman/Milman's converse to show that the active union is norming, and on the coordinate-wise verification of condition (v)' in the ℓ_∞ -sum.

References

- [1] G. Choi and M. Jung, *A generalized ACK structure and the denseness of norm attaining operators*, arXiv:2204.01991.
- [2] R. R. Phelps, *Lectures on Choquet's Theorem*, second edition, Lecture Notes in Mathematics 1757, Springer, 2001.